ETE 4351L Data Communication and Networking
Lab 12: Analyzing TCP/UDP Network Traffic
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OBJECTIVES

- Establish a TCP connection between two computers using SocketTest
- Verify the steps of a TCP handshake
- Capture traffic on a network interface using Wireshark
- Examine encapsulated packets at various OSI layers
- Learn the basics of packet capture software Wireshark

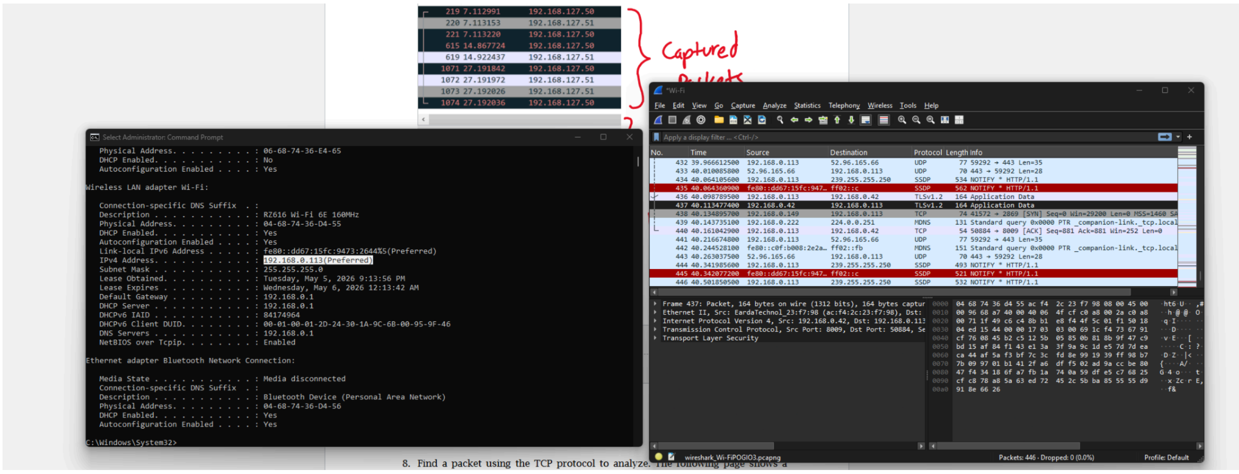
INTRODUCTION

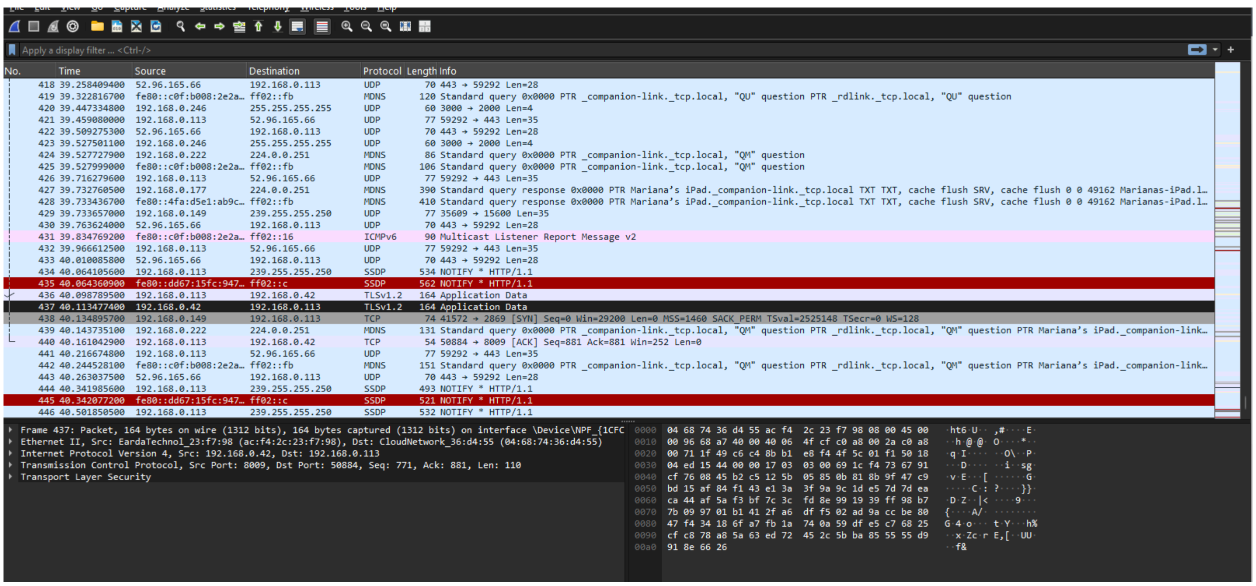
The purpose of this lab was to analyze TCP and UDP network traffic using Wireshark and SocketTest. Wireshark was used to capture packets on the computer Wi-Fi interface while SocketTest was used to generate TCP and UDP traffic between a client and a server. This lab focused on identifying how data is encapsulated at different OSI layers like the ethernet II frame at layer 2, IPv4 packets at layer 3, and the TCP or UDP segment at layer 4.

For this online experiment, we analyzed the TCP traffic to verify connections, data transfers, and connection termination. We also verified packets that show three-way handshaking using SYN, SYN-ACK, and ACK packets followed by FIN/ACKL sequences. We also compared UDP vs. TCP and what happens when we connect to an online server that is not within our local network.

WIRESHARK

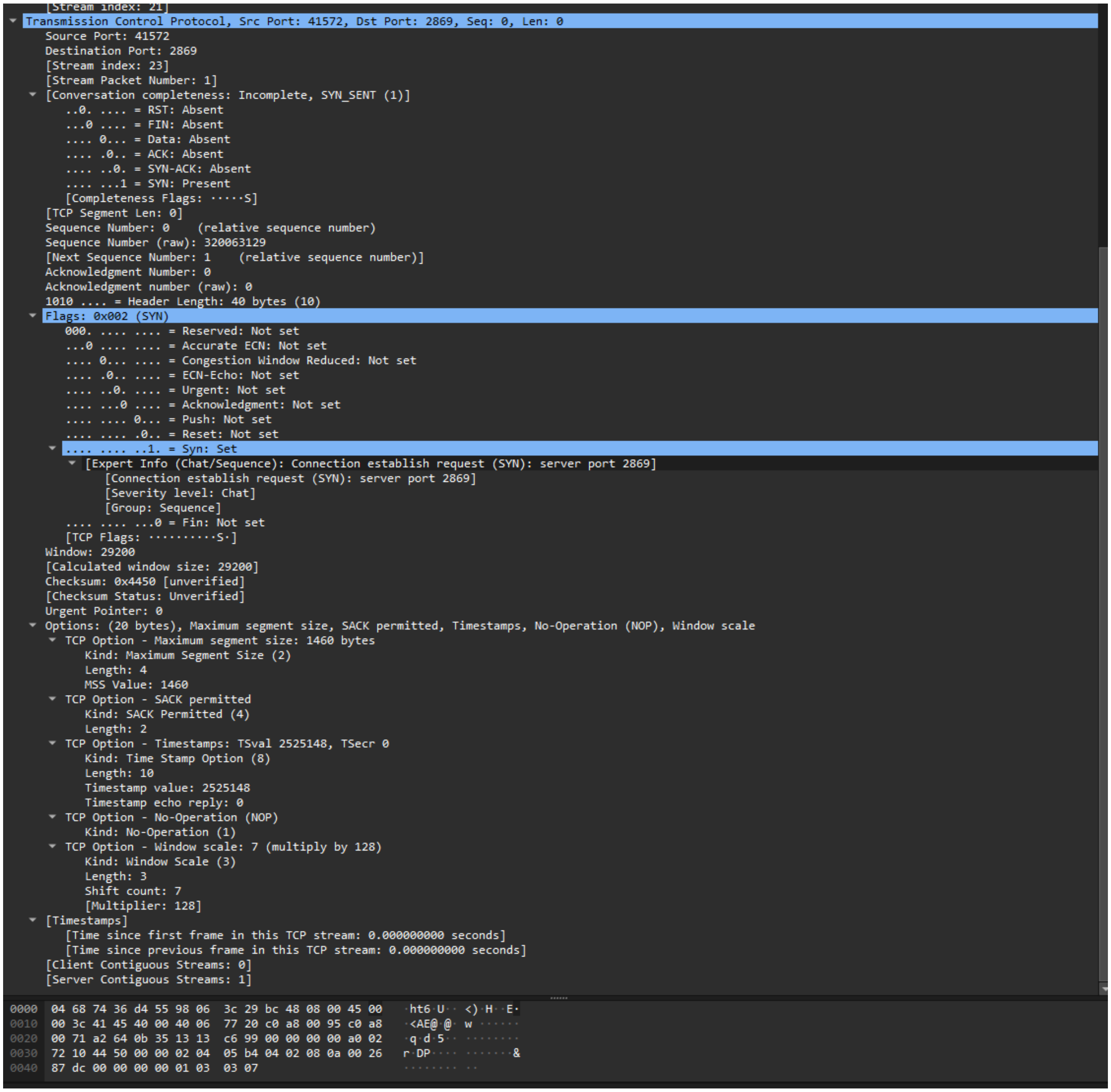
1-7. Setting up WireShark and capturing traffic





8. Find a packet using TCP protocol to analyze. In Wireshark, we see the "Ethernet II" frame, the "Internet Protocol Version 4" packet, and the "Transmission Control Protocol" segment. Each of those can be expanded to show the detailed values captured by Wireshark for each layer. Examine the data captured from your machine to match up with concepts learned in lecture. Show some screenshots to explain what you found and how it relates to lecture materials.

```
Frame 438: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
        Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  5, 2026 22:58:00.942739000 Pacific Daylight Time
    UTC Arrival Time: May  6, 2026 05:58:00.942739000 UTC
    Epoch Arrival Time: 1778047080.942739000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 21.418300 milliseconds]
    [Time delta from previous displayed frame: 21.418300 milliseconds]
    [Time since reference or first frame: 40.134895700 seconds]
    Frame Number: 438
    Frame Length: 74 bytes (592 bits)
    Capture Length: 74 bytes (592 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP SYN/FIN]
    [Coloring Rule String: tcp.flags & 0x02 || tcp.flags.fin == 1]
Ethernet II, Src: SamsungElect_29:bc:48 (98:06:3c:29:bc:48), Dst: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
    Destination: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: SamsungElect_29:bc:48 (98:06:3c:29:bc:48)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 13]
Internet Protocol Version 4, Src: 192.168.0.149, Dst: 192.168.0.113
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 60
    Identification: 0x4145 (16709)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x7720 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.149
    Destination Address: 192.168.0.113
    [Stream index: 21]
Transmission Control Protocol, Src Port: 41572, Dst Port: 2869, Seq: 0, Len: 0
    Source Port: 41572
    Destination Port: 2869
    [Stream index: 23]
    [Stream Packet Number: 1]

0000  04 68 74 36 d4 55 98 06  3c 29 bc 48 08 00 45 00   ·ht6·U·· <)·H··E·
0010  00 3c 41 45 40 00 40 06  77 20 c0 a8 00 95 c0 a8   ·<AE@·@· w ······
0020  00 71 a2 64 0b 35 13 13  c6 99 00 00 00 00 a0 02   ·q·d·5·· ········
0030  72 10 44 50 00 00 02 04  05 b4 04 02 08 0a 00 26   r·DP···· ·······&
```



For this section, the packet is shown as Frame 438 with a total frame size of 74 bytes (592 bits).

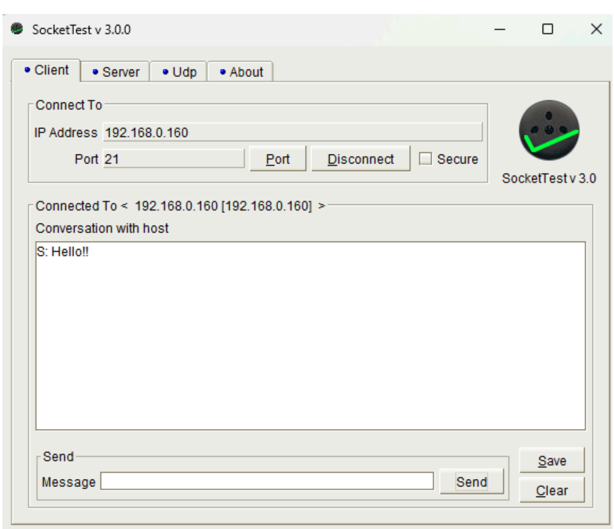
At layer 2, the data link layer, Wireshark shows the packet as an Ethernet II frame and we can see the source MAC address listed as SamsungElect_29:bc:48 (98:06:32:29:bc:48) and the destination MAC address is listed as CloudNetwork_36:d4:55 (04:68:74:36:d4:55). This relates to that OSI layer because ethernet is responsible for local network delivery using MAC addresses. The ethernet frame encapsulates the IP packet before it gets transmitted.

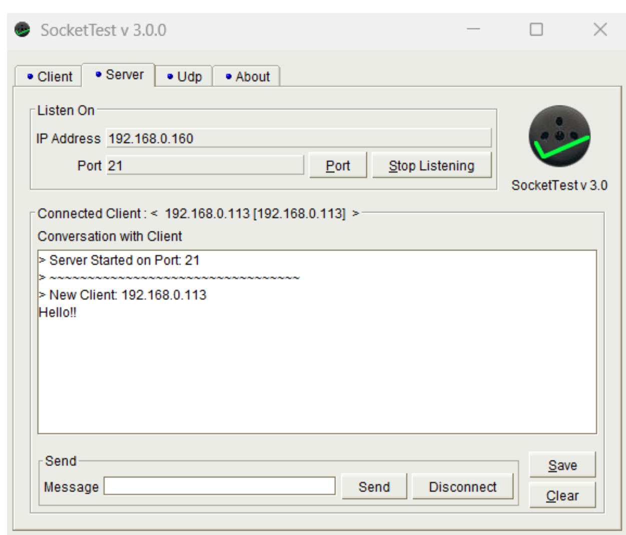
At the network layer (layer 3), the packet now uses Internet Protocol (IP) and the source address is 192.168.0.149 and the destination is 192.168.0.113. The IP header contains information needed to route the packet between devices, including source and destination IP addresses, header length, total length, time to live, and protocol type. We can see the header shows the protocol field is TCP and now the IP packet is carrying a TCP segment.

At the transport layer (layer 4), we can see the source port is 41572 and the destination port is 2869. The most important part of this packet is the TCP flag field, we can see Flags: 0x002 (SYN) and that means the SYN flag is set, ACK flag is not set,it is used to start a TCP connection. The client is requesting to establish a connection with the destination device on port 2869. This coincides with the OSI layer 4 description where the transport layer is responsible for establishing reliable connections between devices before data is transferred.

SOCKET TEST

1-6. The SocketTest program can act as either a TCP client/server, or UDP client/server. In this lab, you will act as the client and establish a connection to the server. Once a connection is established, begin capturing packets on Wireshark. View the packets back and forth for the TCP connection being established, the message, and the TCP connection being terminated.





Verify the contents of the frames by viewing the data at each layer (layer 2, layer 3, layer 4).

• What are the source and destination MAC addresses? Do these values make sense?

Ethernet II Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)

Dst: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)

These values make sense because ethernet uses MAC addresses for local network delivery.

• What other fields do you see as part of the ethernet frame?

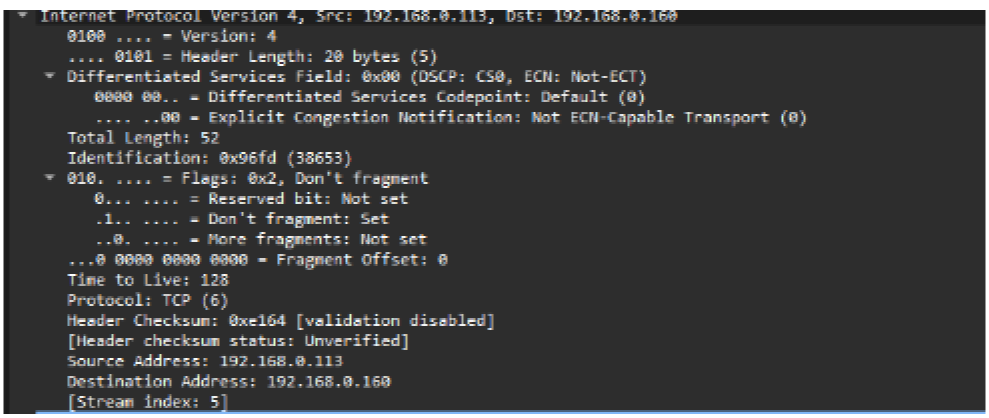
The destination MAC address, the source MAC address and the Type.

• What are the source and destination IP addresses in the IP packet?

Source IP: 192.168.0.113

Destination IP: 192.168.0.160

• Verify the other fields in the IP header

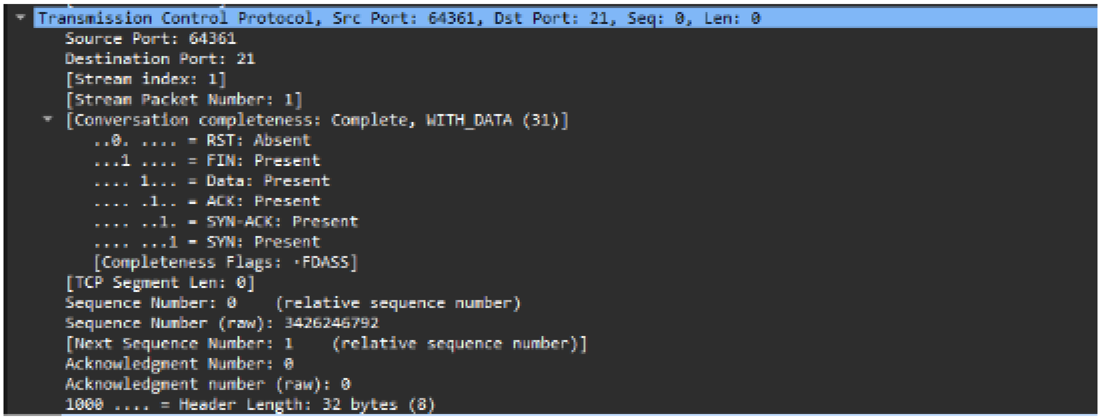


```
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
   0100 .... = Version: 4
   .... 0101 = Header Length: 20 bytes (5)
   Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
   Total Length: 52
   Identification: 0x96fd (38653)
   010. .... = Flags: 0x2, Don't fragment
      0... .... = Reserved bit: Not set
      .1.. .... = Don't fragment: Set
      ..0. .... = More fragments: Not set
   ...0 0000 0000 0000 = Fragment Offset: 0
   Time to Live: 128
   Protocol: TCP (6)
   Header Checksum: 0xe164 [validation disabled]
   [Header checksum status: Unverified]
   Source Address: 192.168.0.113
   Destination Address: 192.168.0.160
   [Stream index: 5]
```

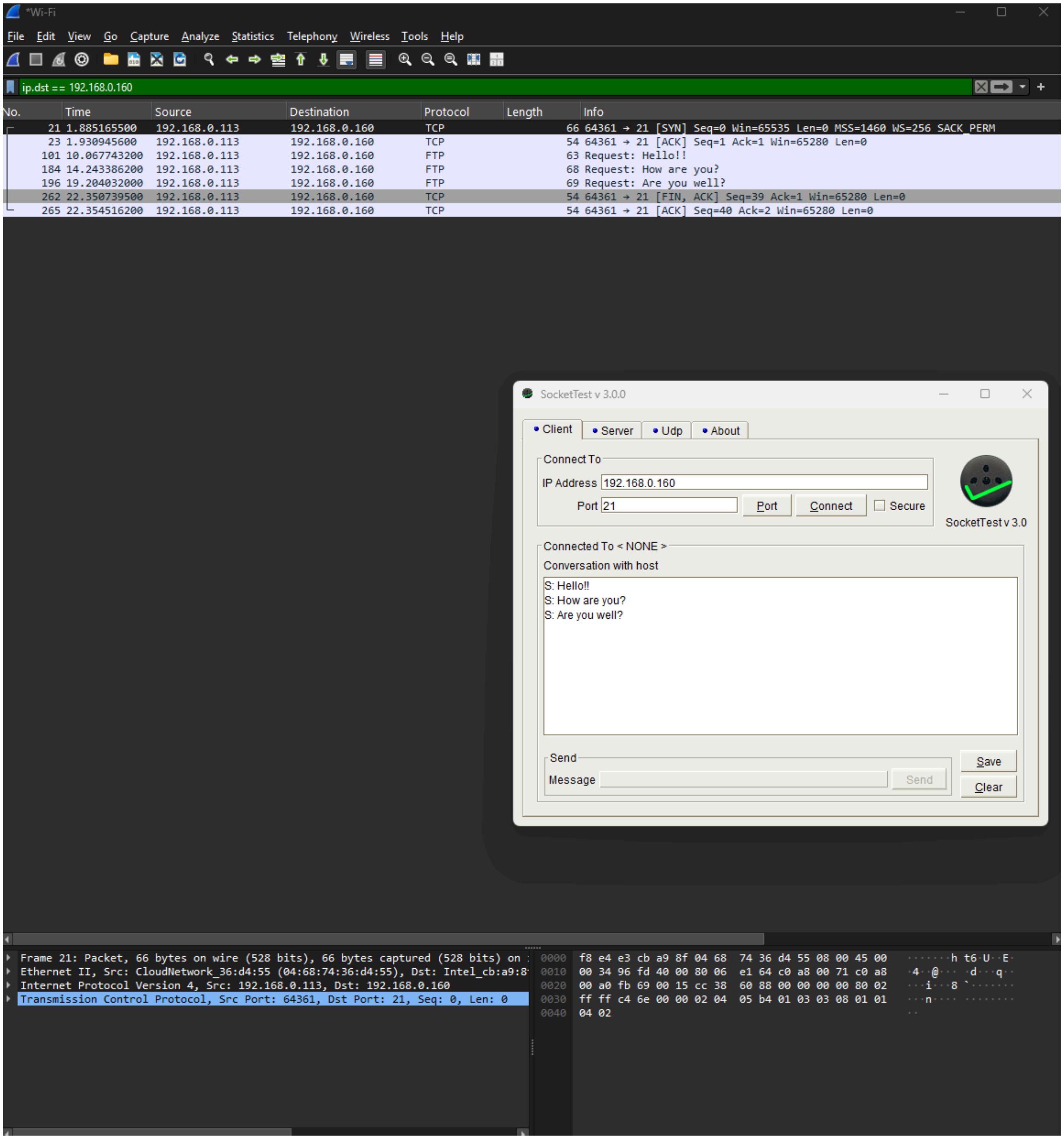
• What are the details of the TCP segment? Which ports are used for Communication?

Source Port: 64361

Destination Port: 21

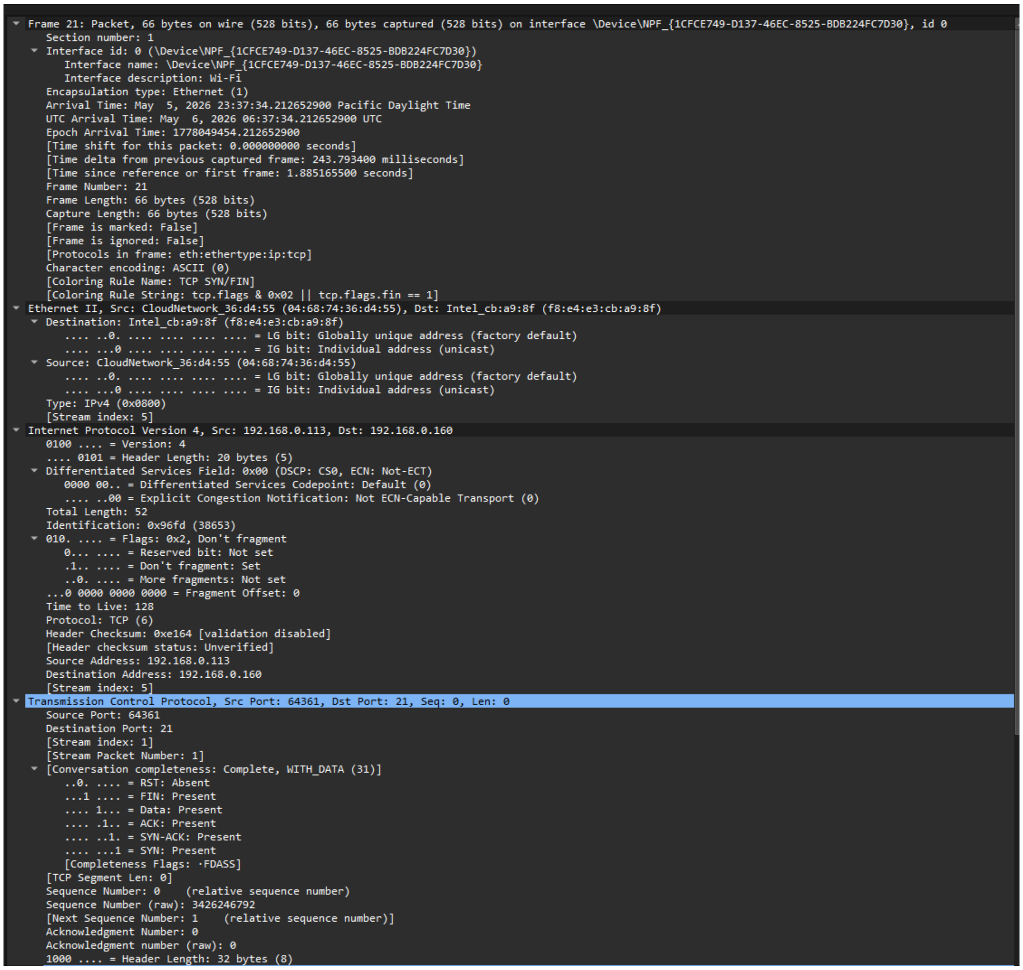


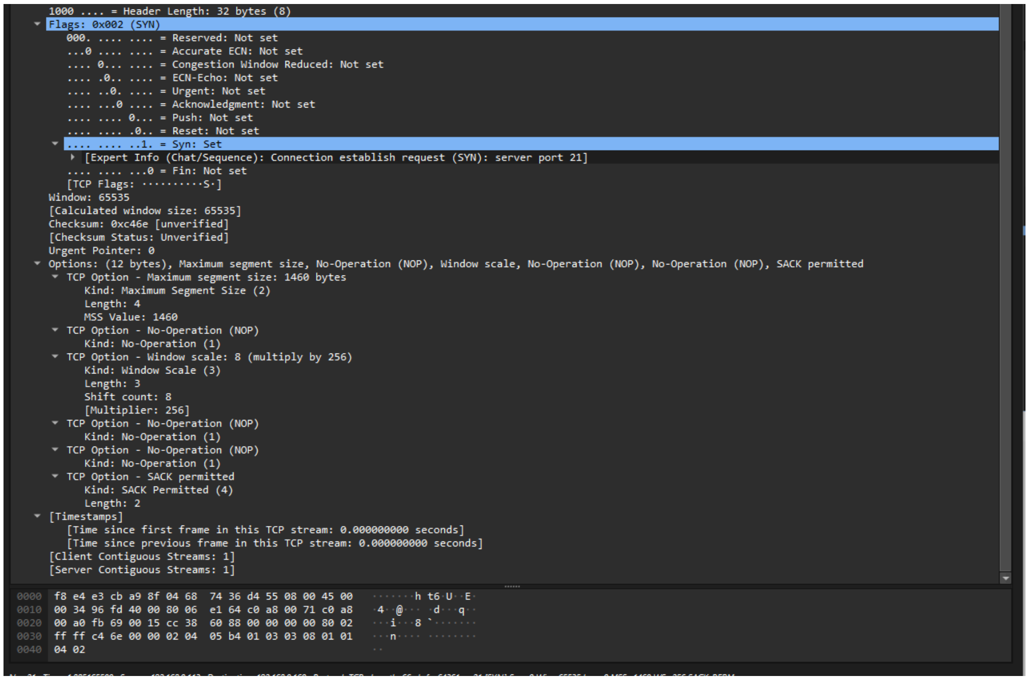
- Verify the SYN, SYN-ACK, ACK behavior.
- Verify the FIN, ACK, FIN, ACK behavior



*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst == 192.168.0.160
No. Time Source Destination Protocol Length Info
21 1.885165500 192.168.0.113 192.168.0.160 TCP 66 64361 → 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
23 1.930945600 192.168.0.113 192.168.0.160 TCP 54 64361 → 21 [ACK] Seq=1 Ack=1 Win=65280 Len=0
101 10.067743200 192.168.0.113 192.168.0.160 FTP 63 Request: Hello!!
184 14.243386200 192.168.0.113 192.168.0.160 FTP 68 Request: How are you?
196 19.204032000 192.168.0.113 192.168.0.160 FTP 69 Request: Are you well?
262 22.350739500 192.168.0.113 192.168.0.160 TCP 54 64361 → 21 [FIN, ACK] Seq=39 Ack=1 Win=65280 Len=0
265 22.354516200 192.168.0.113 192.168.0.160 TCP 54 64361 → 21 [ACK] Seq=40 Ack=2 Win=65280 Len=0
SocketTest v 3.0.0
Client
Server
Udp
About
Connect To
IP Address 192.168.0.160
Port 21
Port
Connect
Secure
SocketTest v 3.0
Connected To < NONE >
Conversation with host
S: Hello!!
S: How are you?
S: Are you well?
Send
Message
Send
Save
Clear
Frame 21: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: Intel_cb:a9:8
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
Transmission Control Protocol, Src Port: 64361, Dst Port: 21, Seq: 0, Len: 0
0000 f8 e4 e3 cb a9 8f 04 68 74 36 d4 55 08 00 45 00
0010 00 34 96 fd 40 00 80 06 e1 64 c0 a8 00 71 c0 a8
0020 00 a0 fb 69 00 15 cc 38 60 88 00 00 00 00 80 02
0030 ff ff c4 6e 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02

Connection TCP: [SYN]




1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... .... = Congestion Window Reduced: Not set
.... .0.. .... = ECN-Echo: Not set
.... ..0. .... = Urgent: Not set
....0 .... = Acknowledgment: Not set
.... 0... = Push: Not set
.... .0.. = Reset: Not set
....1. = Syn: Set
[Expert Info (Chat/Sequence): Connection establish request (SYN): server port 21]
....0 = Fin: Not set
[TCP Flags: ··········S·]
Window: 65535
[Calculated window size: 65535]
Checksum: 0xc46e [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted
TCP Option - Maximum segment size: 1460 bytes
Kind: Maximum Segment Size (2)
Length: 4
MSS Value: 1460
TCP Option - No-Operation (NOP)
Kind: No-Operation (1)
TCP Option - Window scale: 8 (multiply by 256)
Kind: Window Scale (3)
Length: 3
Shift count: 8
[Multiplier: 256]
TCP Option - No-Operation (NOP)
Kind: No-Operation (1)
TCP Option - No-Operation (NOP)
Kind: No-Operation (1)
TCP Option - SACK permitted
Kind: SACK Permitted (4)
Length: 2
[Timestamps]
[Time since first frame in this TCP stream: 0.000000000 seconds]
[Time since previous frame in this TCP stream: 0.000000000 seconds]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
0000 f8 e4 e3 cb a9 8f 04 68 74 36 d4 55 08 00 45 00
0010 00 34 96 fd 40 00 80 06 e1 64 c0 a8 00 71 c0 a8
0020 00 a0 fb 69 00 15 cc 38 60 88 00 00 00 00 80 02
0030 ff ff c4 6e 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02

TCP [ACK]

```
Frame 23: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
        Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  5, 2026 23:37:34.258433000 Pacific Daylight Time
    UTC Arrival Time: May  6, 2026 06:37:34.258433000 UTC
    Epoch Arrival Time: 1778049454.258433000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 193.200 microseconds]
    [Time delta from previous displayed frame: 45.780100 milliseconds]
    [Time since reference or first frame: 1.930945600 seconds]
    Frame Number: 23
    Frame Length: 54 bytes (432 bits)
    Capture Length: 54 bytes (432 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
    Destination: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 5]
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 40
    Identification: 0x96fe (38654)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0xe16f [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.113
    Destination Address: 192.168.0.160
    [Stream index: 5]
Transmission Control Protocol, Src Port: 64361, Dst Port: 21, Seq: 1, Ack: 1, Len: 0
    Source Port: 64361
    Destination Port: 21
    [Stream index: 1]
    [Stream Packet Number: 3]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3426246793
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2624795525
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
```

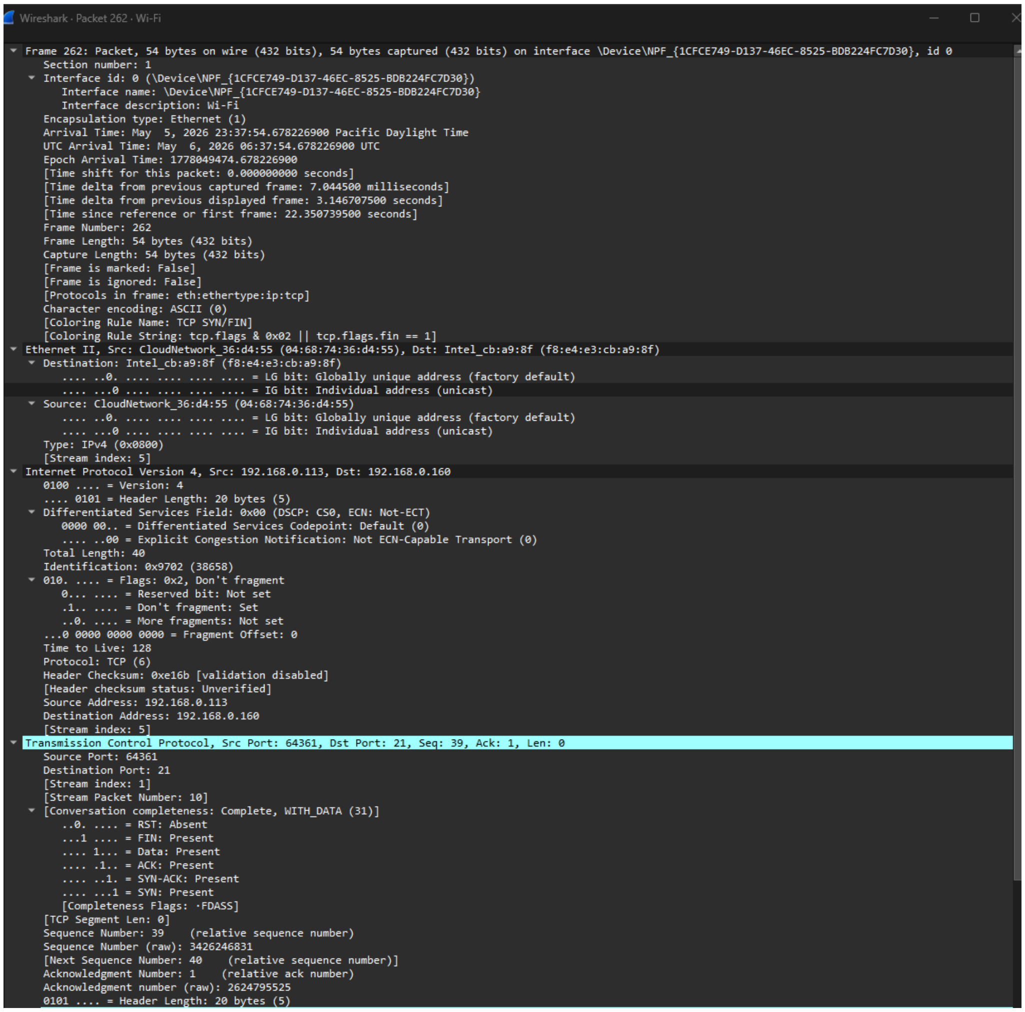
```
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·······A····]
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0x343a [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
        [Time since first frame in this TCP stream: 45.780100 milliseconds]
        [Time since previous frame in this TCP stream: 193.200 microseconds]
    [SEQ/ACK analysis]
        [This is an ACK to the segment in frame: 22]
        [The RTT to ACK the segment was: 193.200 microseconds]
        [iRTT: 45.780100 milliseconds]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]

0000  f8 e4 e3 cb a9 8f 04 68  74 36 d4 55 08 00 45 00   ·······h t6·U··E·
0010  00 28 96 fe 40 00 80 06  e1 6f c0 a8 00 71 c0 a8   ·(··@··· ·o···q··
0020  00 a0 fb 69 00 15 cc 38  60 89 9c 73 33 85 50 10   ···i···8 `··s3·P·
0030  00 ff 34 3a 00 00                                  ··4:··
```

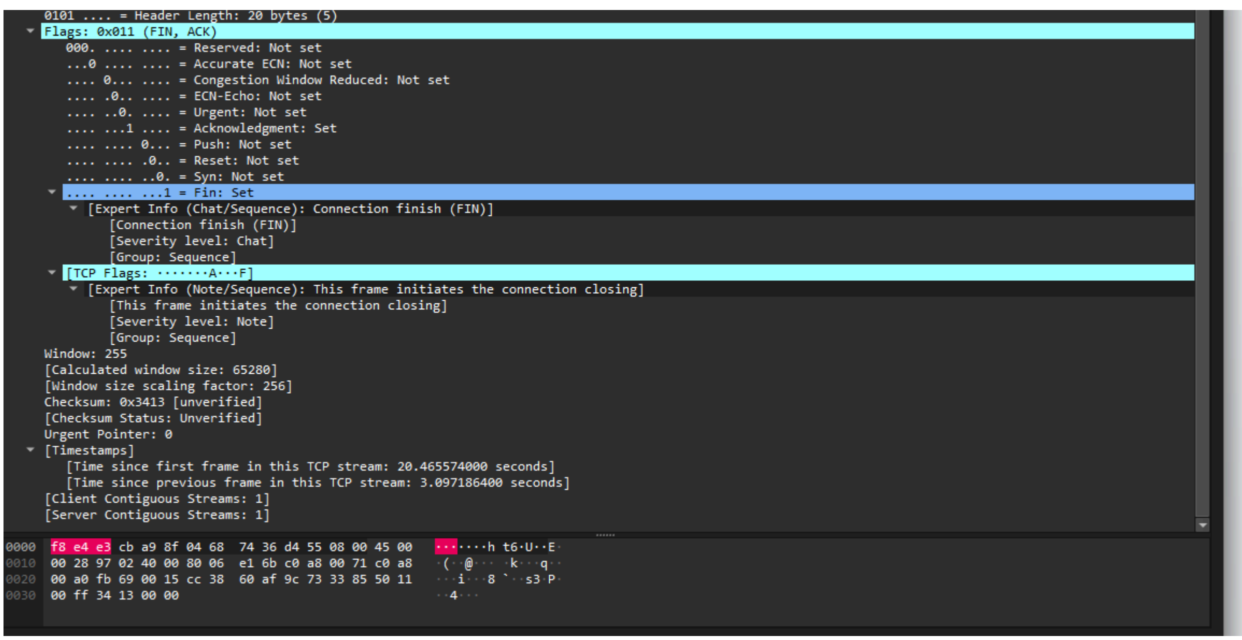
TCP FTP Message:

```
Frame 101: Packet, 63 bytes on wire (504 bits), 63 bytes captured (504 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
        Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  5, 2026 23:37:42.395230600 Pacific Daylight Time
    UTC Arrival Time: May  6, 2026 06:37:42.395230600 UTC
    Epoch Arrival Time: 1778049462.395230600
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 541.512000 milliseconds]
    [Time delta from previous displayed frame: 8.136797600 seconds]
    [Time since reference or first frame: 10.067743200 seconds]
    Frame Number: 101
    Frame Length: 63 bytes (504 bits)
    Capture Length: 63 bytes (504 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:ftp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
    Destination: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 5]
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 49
    Identification: 0x96ff (38655)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0xe165 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.113
    Destination Address: 192.168.0.160
    [Stream index: 5]
Transmission Control Protocol, Src Port: 64361, Dst Port: 21, Seq: 1, Ack: 1, Len: 9
    Source Port: 64361
    Destination Port: 21
    [Stream index: 1]
    [Stream Packet Number: 4]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 9]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3426246793
    [Next Sequence Number: 10    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2624795525
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
```

```
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·······AP···]
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0xe528 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
        [Time since first frame in this TCP stream: 8.182577700 seconds]
        [Time since previous frame in this TCP stream: 8.136797600 seconds]
    [SEQ/ACK analysis]
        [iRTT: 45.780100 milliseconds]
        [Bytes in flight: 9]
        [Bytes sent since last PSH flag: 9]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (9 bytes)
File Transfer Protocol (FTP)
    Hello!!\r\n
    [Current working directory: ]

0000  f8 e4 e3 cb a9 8f 04 68  74 36 d4 55 08 00 45 00   ·······h t6·U··E·
0010  00 31 96 ff 40 00 80 06  e1 65 c0 a8 00 71 c0 a8   ·1··@··· ·e···q··
0020  00 a0 fb 69 00 15 cc 38  60 89 9c 73 33 85 50 18   ···i···8 `··s3·P·
0030  00 ff e5 28 00 00 48 65  6c 6c 6f 21 21 0d 0a      ···(··He llo!!··
```

TCP [FIN ACK]


Wireshark · Packet 262 · Wi-Fi
Frame 262: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
Section number: 1
Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
Interface description: Wi-Fi
Encapsulation type: Ethernet (1)
Arrival Time: May 5, 2026 23:37:54.678226900 Pacific Daylight Time
UTC Arrival Time: May 6, 2026 06:37:54.678226900 UTC
Epoch Arrival Time: 1778049474.678226900
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 7.044500 milliseconds]
[Time delta from previous displayed frame: 3.146707500 seconds]
[Time since reference or first frame: 22.350739500 seconds]
Frame Number: 262
Frame Length: 54 bytes (432 bits)
Capture Length: 54 bytes (432 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP SYN/FIN]
[Coloring Rule String: tcp.flags & 0x02 || tcp.flags.fin == 1]
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
Destination: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 5]
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 40
Identification: 0x9702 (38658)
010. .... = Flags: 0x2, Don't fragment
0... .... = Reserved bit: Not set
.1.. .... = Don't fragment: Set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0xe16b [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.113
Destination Address: 192.168.0.160
[Stream index: 5]
Transmission Control Protocol, Src Port: 64361, Dst Port: 21, Seq: 39, Ack: 1, Len: 0
Source Port: 64361
Destination Port: 21
[Stream index: 1]
[Stream Packet Number: 10]
[Conversation completeness: Complete, WITH_DATA (31)]
..0. .... = RST: Absent
...1 .... = FIN: Present
.... 1... = Data: Present
.... .1.. = ACK: Present
.... ..1. = SYN-ACK: Present
....1 = SYN: Present
[Completeness Flags: ·FDASS]
[TCP Segment Len: 0]
Sequence Number: 39 (relative sequence number)
Sequence Number (raw): 3426246831
[Next Sequence Number: 40 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2624795525
0101 .... = Header Length: 20 bytes (5)

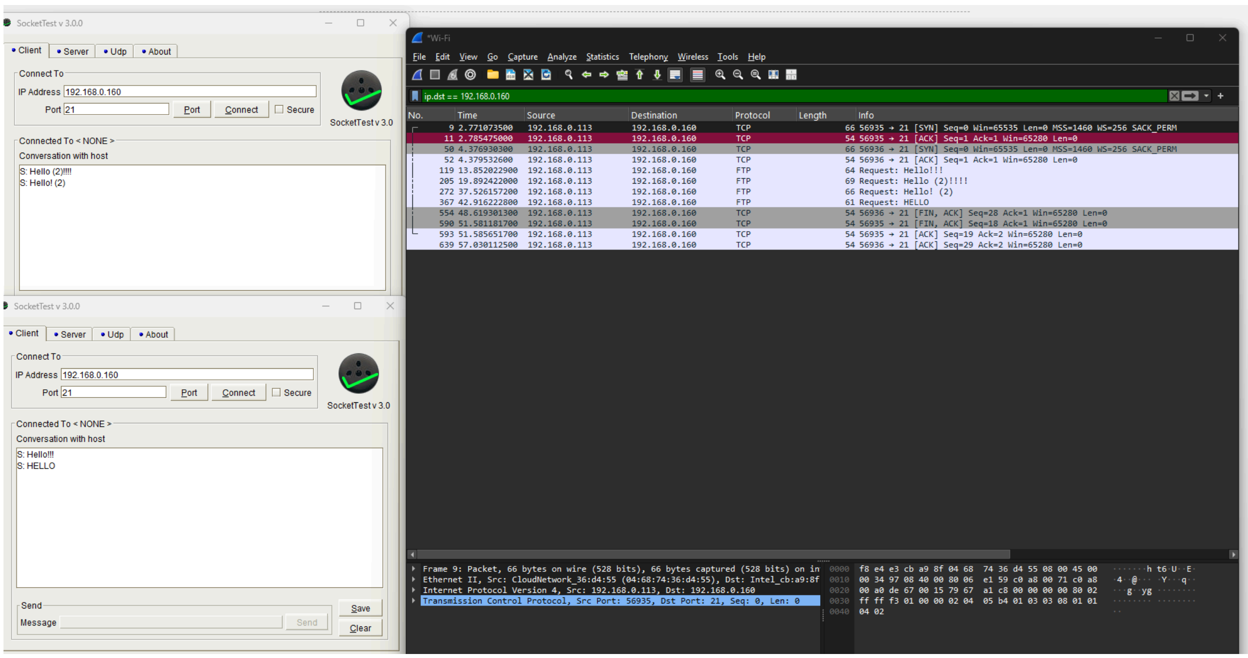

0101 .... = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)
000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... .... = Congestion Window Reduced: Not set
.... .0.. .... = ECN-Echo: Not set
.... ..0. .... = Urgent: Not set
....1 .... = Acknowledgment: Set
.... 0... = Push: Not set
.... .0.. = Reset: Not set
....0. = Syn: Not set
....1 = Fin: Set
[Expert Info (Chat/Sequence): Connection finish (FIN)]
[Connection finish (FIN)]
[Severity level: Chat]
[Group: Sequence]
[TCP Flags: ·······A···F]
[Expert Info (Note/Sequence): This frame initiates the connection closing]
[This frame initiates the connection closing]
[Severity level: Note]
[Group: Sequence]
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x3413 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Time since first frame in this TCP stream: 20.465574000 seconds]
[Time since previous frame in this TCP stream: 3.097186400 seconds]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
0000 f8 e4 e3 cb a9 8f 04 68 74 36 d4 55 08 00 45 00 ·······h t6·U··E·
0010 00 28 97 02 40 00 80 06 e1 6b c0 a8 00 71 c0 a8 ·(··@··· ·k···q··
0020 00 a0 fb 69 00 15 cc 38 60 af 9c 73 33 85 50 11 ···i···8 `··s3·P·
0030 00 ff 34 13 00 00 ··4···

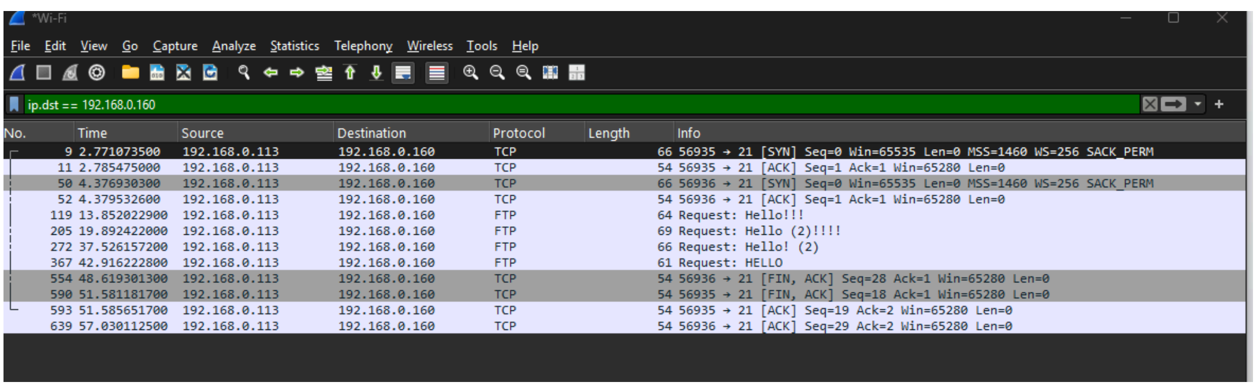
TCP [ACK] Disconnection:

```
Frame 265: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
        Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  5, 2026 23:37:54.682003600 Pacific Daylight Time
    UTC Arrival Time: May  6, 2026 06:37:54.682003600 UTC
    Epoch Arrival Time: 1778049474.682003600
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 145.700 microseconds]
    [Time delta from previous displayed frame: 3.776700 milliseconds]
    [Time since reference or first frame: 22.354516200 seconds]
    Frame Number: 265
    Frame Length: 54 bytes (432 bits)
    Capture Length: 54 bytes (432 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
    Destination: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 5]
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 40
    Identification: 0x9703 (38659)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0xe16a [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.113
    Destination Address: 192.168.0.160
    [Stream index: 5]
Transmission Control Protocol, Src Port: 64361, Dst Port: 21, Seq: 40, Ack: 2, Len: 0
    Source Port: 64361
    Destination Port: 21
    [Stream index: 1]
    [Stream Packet Number: 13]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 0]
    Sequence Number: 40    (relative sequence number)
    Sequence Number (raw): 3426246832
    [Next Sequence Number: 40    (relative sequence number)]
    Acknowledgment Number: 2    (relative ack number)
    Acknowledgment number (raw): 2624795526
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
```

```
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·······A····]
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0x3412 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
        [Time since first frame in this TCP stream: 20.469350700 seconds]
        [Time since previous frame in this TCP stream: 145.700 microseconds]
    [SEQ/ACK analysis]
        [This is an ACK to the segment in frame: 264]
        [The RTT to ACK the segment was: 145.700 microseconds]
        [iRTT: 45.780100 milliseconds]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]

0000  f8 e4 e3 cb a9 8f 04 68  74 36 d4 55 08 00 45 00   ·······h t6·U··E·
0010  00 28 97 03 40 00 80 06  e1 6a c0 a8 00 71 c0 a8   ·(··@··· ·j···q··
0020  00 a0 fb 69 00 15 cc 38  60 b0 9c 73 33 86 50 10   ···i···8 `··s3·P·
0030  00 ff 34 12 00 00                                  ··4···
```

7. This time, open up two instances of the SocketTest program on your computer. Use each one at the same time to establish two simultaneous connections to the server. Repeat the steps of establishing connections, sending messages, and closing connections. View the packets back and forth for the TCP connection being established, the message, and the TCP connection being terminated.





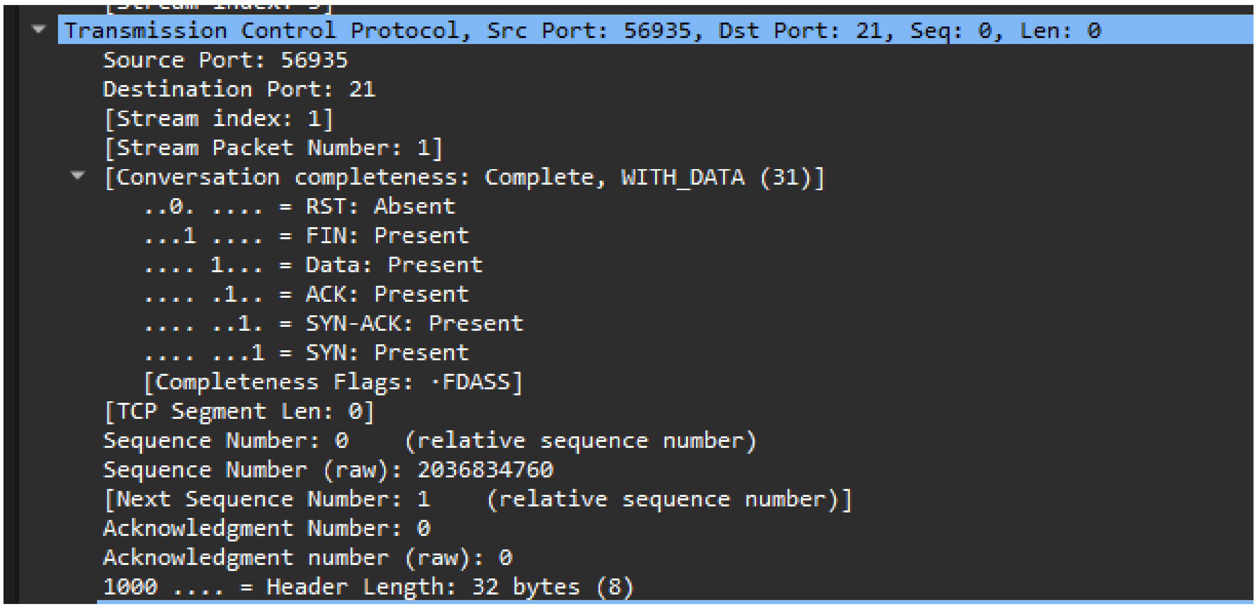
Briefly compare the two different TCP connections by analyzing the Wireshark capture.

Both connections used the same client IP address, 192.168.0.113, the same server IP address, 192.168.0.160 and the same destination port, 21. But, they used different client source ports, one used source port 56935 and the other used source port 56936.

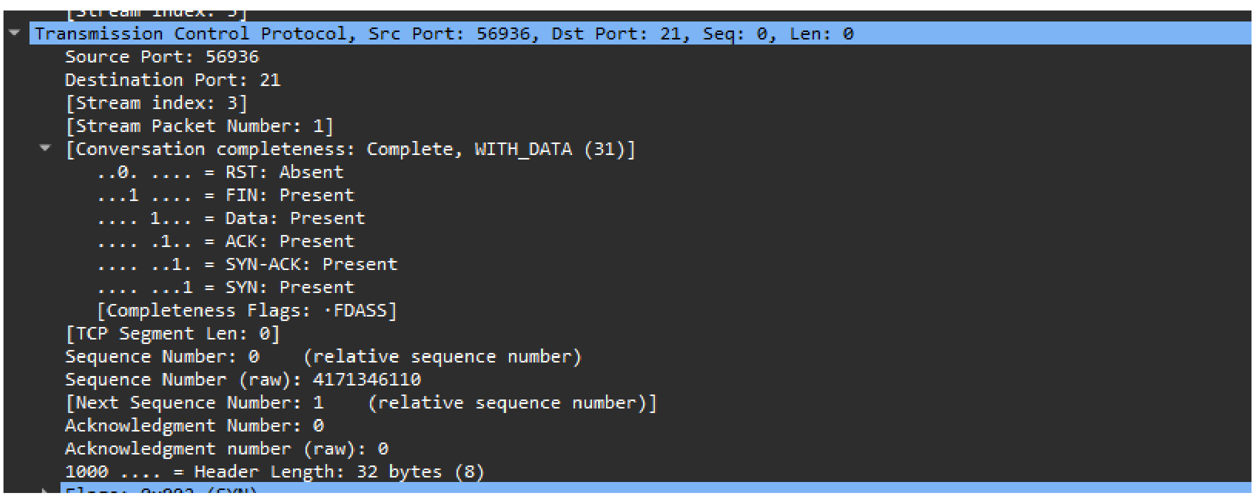
How does TCP distinguish between the two connections?

TCP distinguishes between the two connections by using the combination of the source IP, source port, destination IP, destination port, and the protocol. Since the source ports are different, the TCP connection treats them as two separate connections even though they are communicating with the same exact server.

The two TCP [SYN] for the two connections:



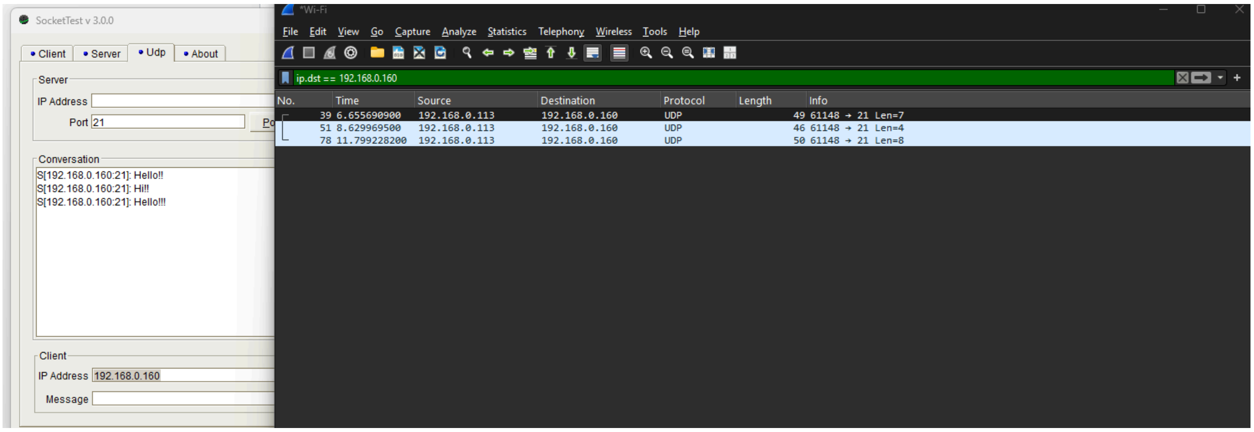
```
Transmission Control Protocol, Src Port: 56935, Dst Port: 21, Seq: 0, Len: 0
    Source Port: 56935
    Destination Port: 21
    [Stream index: 1]
    [Stream Packet Number: 1]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 2036834760
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
```



```
Transmission Control Protocol, Src Port: 56936, Dst Port: 21, Seq: 0, Len: 0
    Source Port: 56936
    Destination Port: 21
    [Stream index: 3]
    [Stream Packet Number: 1]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 4171346110
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
```

8. Now, use SocketTest to send traffic using UDP instead of TCP. Remember that UDP is a connection-less protocol. So, there is no handshaking or establishing communication link before sending data. Here, you can switch to the UDP tab and then use the bottom portion of the program to enter the client's IP address and

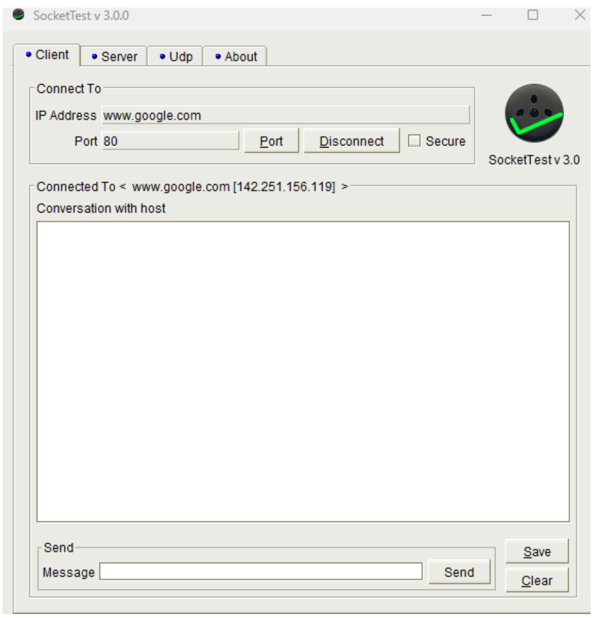
send data. Examine the traffic using Wireshark. Consult the UDP segment protocol for comparison.

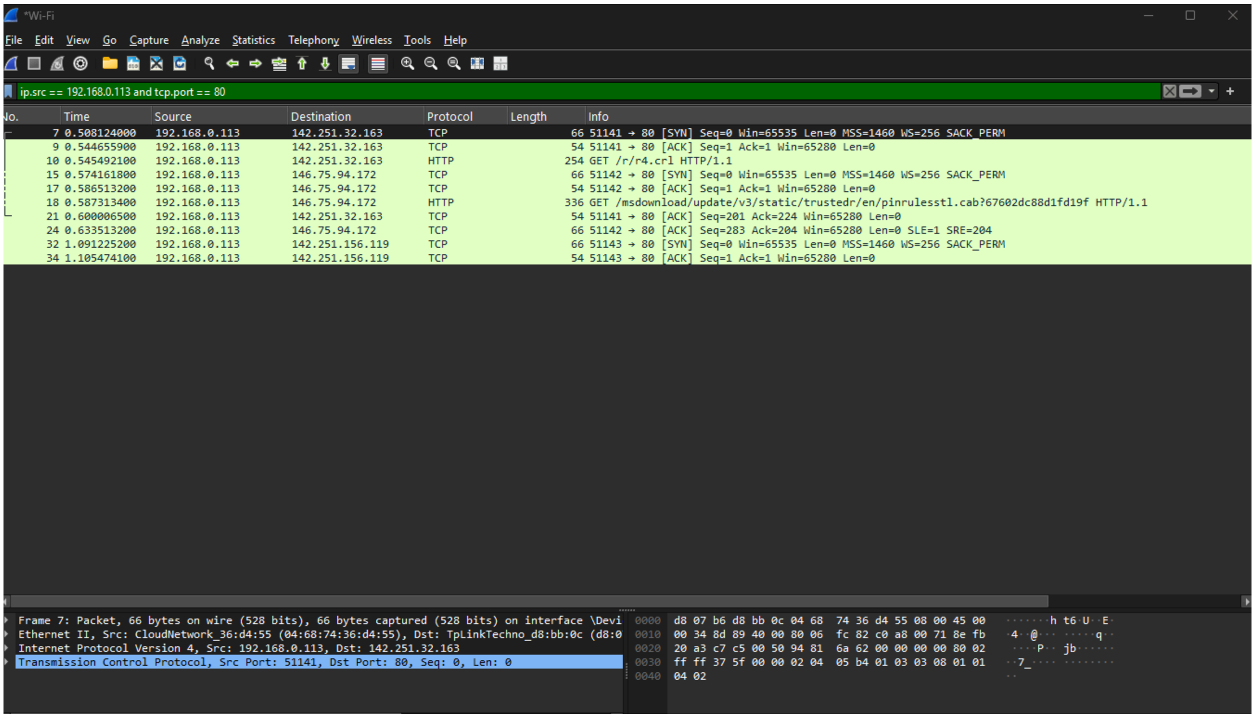


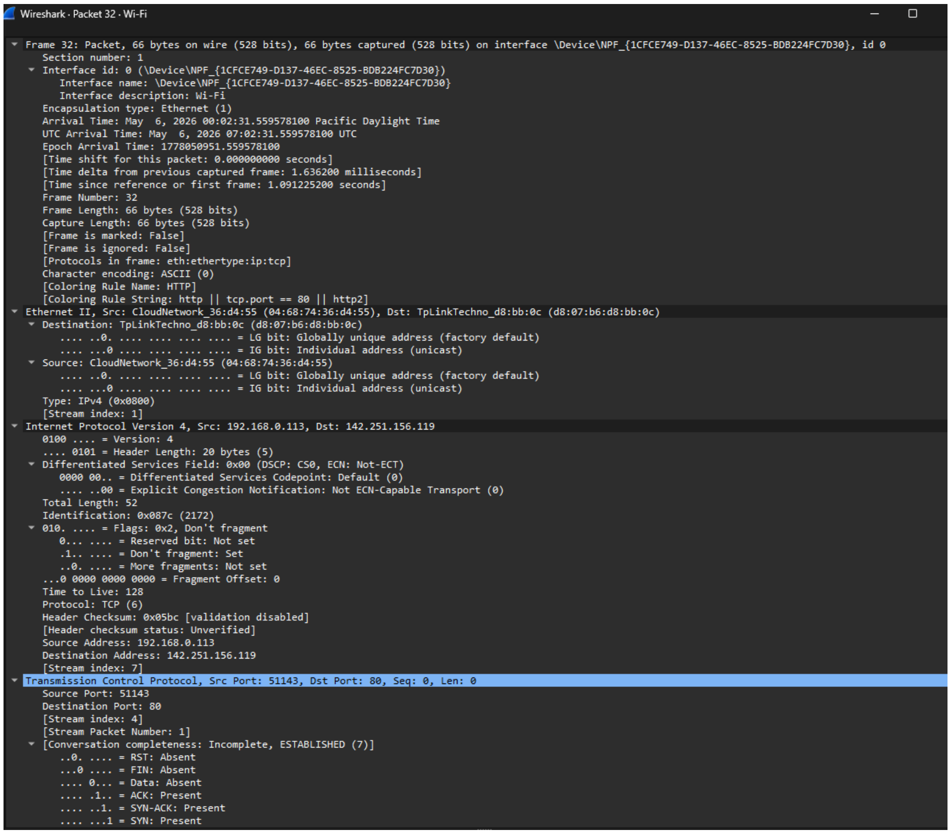
```
Frame 39: Packet, 49 bytes on wire (392 bits), 49 bytes captured (392 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
        Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  5, 2026 23:57:36.145158100 Pacific Daylight Time
    UTC Arrival Time: May  6, 2026 06:57:36.145158100 UTC
    Epoch Arrival Time: 1778050656.145158100
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 86.690500 milliseconds]
    [Time since reference or first frame: 6.655690900 seconds]
    Frame Number: 39
    Frame Length: 49 bytes (392 bits)
    Capture Length: 49 bytes (392 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:udp:data]
    Character encoding: ASCII (0)
    [Coloring Rule Name: UDP]
    [Coloring Rule String: udp]
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
    Destination: Intel_cb:a9:8f (f8:e4:e3:cb:a9:8f)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 5]
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 192.168.0.160
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 35
    Identification: 0x9714 (38676)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x2154 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.113
    Destination Address: 192.168.0.160
    [Stream index: 11]
User Datagram Protocol, Src Port: 61148, Dst Port: 21
    Source Port: 61148
    Destination Port: 21
    Length: 15
    Checksum: 0x4989 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 4]
    [Stream Packet Number: 1]
    [Timestamps]
        [Time since first frame: 0.000000000 seconds]
        [Time since previous frame: 0.000000000 seconds]
    UDP payload (7 bytes)
Data (7 bytes)
    Data: 48656c6c6f2121
    [Length: 7]

0000  f8 e4 e3 cb a9 8f 04 68  74 36 d4 55 08 00 45 00   ·······h t6·U··E·
0010  00 23 97 14 00 00 80 11  21 54 c0 a8 00 71 c0 a8   ·#······ !T···q··
0020  00 a0 ee dc 00 15 00 0f  49 89 48 65 6c 6c 6f 21   ········ I·Hello!
0030  21                                                 !
```

For this segment, UDP was used instead of TCP. Unlike TCP, there was no SYN, SYN-ACK, ACK handshake before sending data and no FIN/ACK termination sequence afterward. This verifies that UDP is connectionless and sends data without the need to establish a reliable connection.

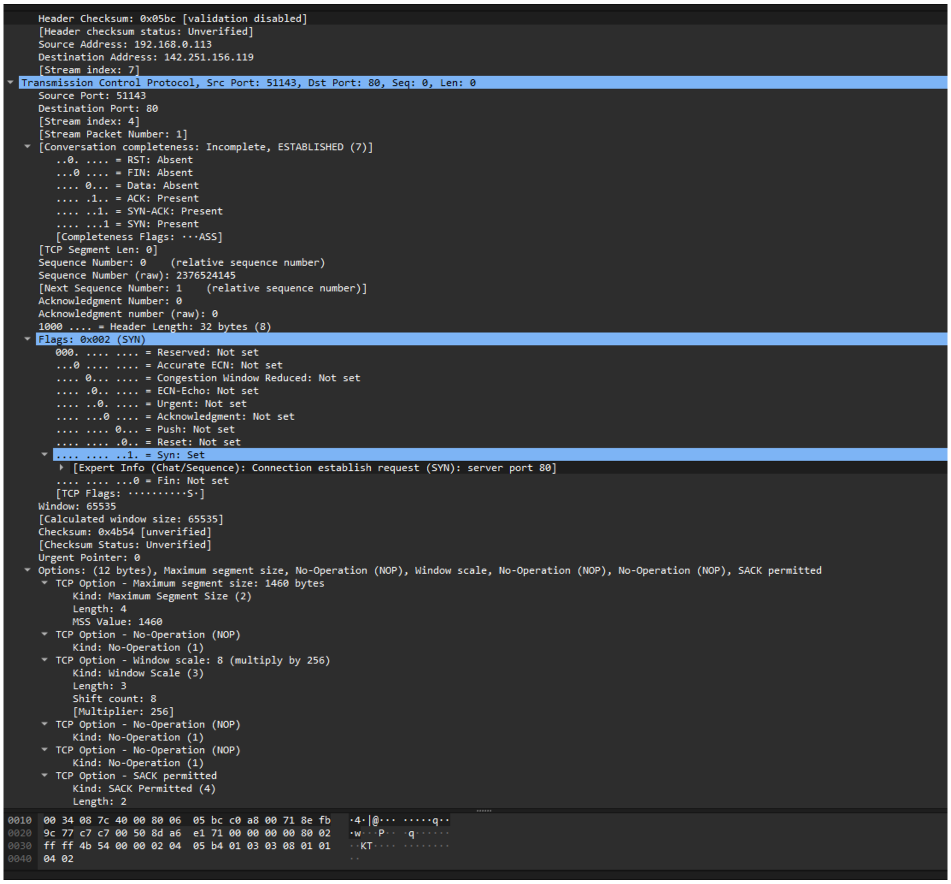
9. Finally, use SocketTest to establish a TCP connection to IP address: www.google.com and port 80. Use Wireshark filters to filter traffic only to your IP address as the source and tcp.port == 80. This will allow you to determine the IP address of the Google server you are connected to. Once you know this, you can add it to your filters as before so that you can see traffic both to and from the server to your machine. Analyze the traffic to determine src/dst IP and MAC addresses, etc. In this case, the destination of the traffic is not in our local subnet (subnetwork), how does this affect the packet data? What is the destination MAC address for the traffic from your machine?




*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src == 192.168.0.113 and tcp.port == 80
No. Time Source Destination Protocol Length Info
7 0.508124000 192.168.0.113 142.251.32.163 TCP 66 51141 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9 0.544655900 192.168.0.113 142.251.32.163 TCP 54 51141 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
10 0.545492100 192.168.0.113 142.251.32.163 HTTP 254 GET /r/r4.crl HTTP/1.1
15 0.574161800 192.168.0.113 146.75.94.172 TCP 66 51142 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
17 0.586513200 192.168.0.113 146.75.94.172 TCP 54 51142 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
18 0.587313400 192.168.0.113 146.75.94.172 HTTP 336 GET /msdownload/update/v3/static/trustedr/en/pinrulesstl.cab?67602dc88d1fd19f HTTP/1.1
21 0.600006500 192.168.0.113 142.251.32.163 TCP 54 51141 → 80 [ACK] Seq=201 Ack=224 Win=65280 Len=0
24 0.633513200 192.168.0.113 146.75.94.172 TCP 66 51142 → 80 [ACK] Seq=283 Ack=204 Win=65280 Len=0 SLE=1 SRE=204
32 1.091225200 192.168.0.113 142.251.156.119 TCP 66 51143 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
34 1.105474100 192.168.0.113 142.251.156.119 TCP 54 51143 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
Frame 7: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Devi
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: TpLinkTechno_d8:bb:0c (d8:0
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 142.251.32.163
Transmission Control Protocol, Src Port: 51141, Dst Port: 80, Seq: 0, Len: 0
0000 d8 07 b6 d8 bb 0c 04 68 74 36 d4 55 08 00 45 00
0010 00 34 8d 89 40 00 80 06 fc 82 c0 a8 00 71 8e fb
0020 20 a3 c7 c5 00 50 94 81 6a 62 00 00 00 00 80 02
0030 ff ff 37 5f 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02



```
Wireshark · Packet 32 · Wi-Fi
Frame 32: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30})
        Interface name: \Device\NPF_{1CFCE749-D137-46EC-8525-BDB224FC7D30}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: May  6, 2026 00:02:31.559578100 Pacific Daylight Time
    UTC Arrival Time: May  6, 2026 07:02:31.559578100 UTC
    Epoch Arrival Time: 1778050951.559578100
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 1.636200 milliseconds]
    [Time since reference or first frame: 1.091225200 seconds]
    Frame Number: 32
    Frame Length: 66 bytes (528 bits)
    Capture Length: 66 bytes (528 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Ethernet II, Src: CloudNetwork_36:d4:55 (04:68:74:36:d4:55), Dst: TpLinkTechno_d8:bb:0c (d8:07:b6:d8:bb:0c)
    Destination: TpLinkTechno_d8:bb:0c (d8:07:b6:d8:bb:0c)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: CloudNetwork_36:d4:55 (04:68:74:36:d4:55)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 1]
Internet Protocol Version 4, Src: 192.168.0.113, Dst: 142.251.156.119
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 52
    Identification: 0x087c (2172)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x05bc [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.113
    Destination Address: 142.251.156.119
    [Stream index: 7]
Transmission Control Protocol, Src Port: 51143, Dst Port: 80, Seq: 0, Len: 0
    Source Port: 51143
    Destination Port: 80
    [Stream index: 4]
    [Stream Packet Number: 1]
    [Conversation completeness: Incomplete, ESTABLISHED (7)]
        ..0. .... = RST: Absent
        ...0 .... = FIN: Absent
        .... 0... = Data: Absent
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
```



When we connected to google.com on port 80, SocketTest had the server IP address to 142.251.156.119. In Wireshark, we saw the IPv4 packet showed a source IP of 192.168.0.113 and destination IP 142.251.156.119. The TCP segment used source port 51143 and destination port 80 with the SYN flag set, starting the handshake. The source MAC address was 04:68:74:36:d4:55 while the destination was d8:07:b6:d8:bb:0c.

Since Google's server is not on the local subnet, the destination MAC address is the router/default gateway rather than the Google server. Google.com is outside the local subnet and our computer cannot send the ethernet frame directly to the MAC address so it has to send to the default gateway/router first and then it can forward the packet outside the local network.

CONCLUSION

This lab helped us understand and demonstrate how network communication is organized through encapsulation across multiple OSI layers, especially layers 2, 3, and 4. The TCP

Wireshark captures verified how TCP is connection oriented and how it completed a three-way handshake using SYN, SYN-ACK, and ACK packets. After messages were sent, we saw the FIN and ACK packets. When we tried to connect to the same server by simultaneous TCP connections to the same server, Wireshark showed that TCP distinguished the connections by assigning different client source ports. When we used UDP, we saw a simpler process where unlike TCP, it did not need to handshake, or use SYN, SYN-ACK, and ACK. We did not even see FIN or ACK. This showed that UDP was simpler and did not need to set up a reliable connection to send data. Finally, when we connected to Google, we saw that the IP packet kept the remote server's IP address as the destination but the ethernet frame uses the router/default gateway's MAC address as the next destination instead since Google is not on the local network so it needs to send data to the router/default and then it routes that to the remote server. Overall, the lab was interesting and we learned a lot about how network communication works as indicated by the OSI model.